

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – FinancialGuide AI

Domain of Project – Artificial Intelligence (AI) & Generative AI

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

Problem Statement : AI Agent for Digital Financial Literacy

The Challenge

An AI Agent for Digital Financial Literacy, powered by RAG (Retrieval-Augmented Generation), helps users understand and navigate essential financial tools and practices. It retrieves reliable content on using UPI, avoiding online scams, understanding interest rates, budgeting, and personal finance management from government portals, banking websites, and educational platforms. With multilingual support, users from diverse backgrounds can interact in their preferred language and ask questions like “How do I send money via UPI?” or “What is a safe interest rate for a loan?” The agent ensures financial literacy is accessible, personalized, and culturally inclusive. This AI-driven assistant empowers users with knowledge, protects them from fraud, and builds confidence in digital finance.

The Objective

- Develop an AI-powered digital financial literacy assistant using IBM watsonx Orchestrate.
- Provide accurate, trustworthy, and easy-to-understand financial guidance using Retrieval-Augmented Generation (RAG).
- Help users understand digital banking services such as UPI, digital payments, online banking, and mobile wallets.
- Educate users about budgeting, saving, investments, loans, interest rates, and credit scores. Increase awareness of online financial scams, phishing attacks, and cybersecurity best practices.
- Deliver responses based on reliable sources such as RBI, NPCI, government portals, and trusted banking resources. Enable natural language interaction so users can ask financial questions in a conversational manner.
- Improve financial decision-making by providing personalized educational information and practical guidance.

Proposed Solution -

Proposed Solution: AI Agent for Digital Financial Literacy

FinancialGuide AI is an AI-powered digital financial literacy assistant developed using **IBM watsonx Orchestrate** and **Retrieval-Augmented Generation (RAG)**. It helps users understand essential financial concepts through a conversational interface by providing reliable, easy-to-understand, and context-aware responses.

The system retrieves information from trusted knowledge sources such as **RBI, NPCI, SEBI, and NCFE**, ensuring that users receive accurate guidance on topics including **UPI, digital payments, budgeting, savings, investments, loans, credit scores, insurance, and online financial fraud prevention**. By combining IBM foundation models with a verified knowledge base, the assistant minimizes misinformation and improves the quality of financial guidance.

To make financial learning more practical, the agent also supports **PDF guide generation** and **Excel-based budget templates**, enabling users to apply financial concepts in everyday life. The solution is designed to be **scalable, secure, and deployment-ready**, allowing future integration of multilingual support, additional financial services, and advanced AI capabilities to promote financial inclusion and digital literacy.

Technology Used

- **IBM watsonx Orchestrate** – Used to build and deploy the AI agent with conversational workflows.
- **IBM Granite Foundation Model** – Provides natural language understanding, reasoning, and intelligent response generation.
- **IBM Cloud** – Offers a secure and scalable platform for hosting and deploying the solution.
- **Retrieval-Augmented Generation (RAG)** – Retrieves relevant information from trusted financial documents before generating responses.
- **IBM watsonx.ai** – Provides access to foundation models and AI services used by the agent.
- **Knowledge Base (RBI, NPCI, SEBI, NCFE)** – Supplies reliable financial information for accurate and trustworthy answers.
- **Built-in Tools (Generate PDF & Create Excel)** – Generates downloadable financial guides and budget planning spreadsheets.

Orchestrate component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized fitness insights and recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time exercise recommendations
- 4) **Chatoutput** -Displays AI-generated responses such as diet plans, fitness analysis, and health suggestions.
- 5) **File component** - Stores and retrieves user data, daily exercise routine, and workout plans for analysis and tracking.

Agent Instructions Screenshot :

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

Home / Manage agents / FinancialGuide_AI

Test agentDeploy

FinancialGuide_AI

Model : GPT-OSS 120B — Ope...

Build with AITalk to agent

Running in debug mode

ProfileKnowledgeToolsetBehaviorChannels Preview

Profile

Describe what your agent does so people and other agents know when to use it.

Description*

- If unsure, recommend consulting an official bank or government website.

Example questions:

- What is SIP?
- How does UPI work?
- How can I avoid online scams?
- What is a credit score?
- Explain EMI with an example.

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

What makes a good description?

Welcome message

Customize the welcome message from the agent on the home screen.

Welcome message

54/100

Hi! I'm FinancialGuide AI.

Your Financial Advisor.

Maximum 100 characters allowed in welcome message edit field.

9:29 PM

Hi! I'm FinancialGuide AI.

Your Financial Advisor.

Accuracy of generated answers may vary. Please double-check responses.

Calculate EMI.

Difference between SIP and FD.

Explain credit score.

< 1/2 >

Explain the difference between NEFT, RTGS, IMPS, and UPI.

Prompt 1 Screenshot :

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

Home / Manage agents / FinancialGuide_AI

Test agentDeploy

FinancialGuide_AI

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Today

You 9:30 PM

Explain the difference between NEFT, RTGS, IMPS, and UPI.

FinancialGuide_AI 9:30 PM

NEFT, RTGS, IMPS, and UPI – what's the difference?

Feature	NEFT (National Electronic Funds Transfer)	RTGS (Real-Time Gross Settlement)	IMPS (Immediate Payment Service)	UPI (Unified Payments Interface)
Purpose	Transfer of funds between banks (usually for larger amounts)	Same-day high-value inter-bank transfers	Instant person-to-person or person-to-merchant transfers, even for small amounts	Mobile-first app-based payments that can work anywhere (bank-to-merchant, etc.)

Save as test

Type something...

Prompt 1 Answer Screenshot :

The screenshot displays the IBM watsonx Orchestrate web application. The top navigation bar includes links for Home, Manage agents, and FinancialGuide_AI. The main header shows the agent name "FinancialGuide_AI" and its model "GPT-OSS 120B - Ope...".

Left Sidebar:

- Profile:** Contains a description field with example questions like "What is SIP?", "How does UPI work?", etc.
- Knowledge:** A section for adding knowledge to the agent.
- Toolset:** A section for defining tools available to the agent.
- Behavior:** A section for configuring agent behavior.
- Channels:** A section for selecting communication channels, currently showing "Preview".

Main Content Area:

- Build with AI / Talk to agent:** Two tabs at the top of the right panel.
- Running in debug mode:** A notification banner.
- Settlement speed:** A table comparing different settlement methods:

Method	Description
Hourly batches	(now 24 x 7). Money reaches the beneficiary within a few hours, sometimes the next day for off-peak slots.
Real-time, individual settlement	funds are transferred instantly, 24 x 7 (since 2022).
Instant	usually a few seconds, 24 x 7.
Instant - typically a few seconds	typically a few seconds, 24 x 7.
- Transaction value limits:** A table comparing transaction limits:

Limit Type	Details
No upper limit; lower limit often ₹1,000	lower limit often ₹1,000 (set by banks).
Minimum ₹2 lakhs	₹200,000); no upper limit.
Maximum ₹5 lakhs	maximum ₹5 lakhs per transaction (some banks allow up to ₹5 lakhs).

Bottom Section:

- Welcome message:** A field to customize the welcome message from the agent on the home screen. Example: "Hi! I'm FinancialGuide AI. Your Financial Advisor."

Prompt 1 Answer Screenshot :

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days leftKD

Home / Manage agents / FinancialGuide_AI

Test agentDeploy

FinancialGuide_AI

Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels Preview

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How it works

sender

some banks may charge a nominal fee.

may charge a small fee for very high-value transfers.

levy a small fee for large volumes.

may pay a small fee (often $\leq 0.5\%$).

How it works

- You give the beneficiary's **bank account number, IFSC, and amount.**

- The system batches transactions and settles them at set intervals.

- Uses the **NEFT clearing house** managed by RBI.

- You give the same details as NEFT.

- Each transaction is settled **individually** in real time, not batched.

- Managed by RBI's **RTGS system.**

- You can use **mobile number, Aadhaar, or virtual ID (VPA)** plus the amount.

- Works via the **National Payments Corporation of India (NPCI)** network.

- You use a **Virtual Payment Address (VPA)** (e.g., `alice@bank1`) (e.g., QR code, or bank account details).

- The app contacts the **NPCI's UP server**, which routes the request to respective banks instantly.

Save as test

Type something...

Prompt 2 Answer Screenshot :

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days leftKD

Home / Manage agents / FinancialGuide_AI

Test agentDeploy

FinancialGuide_AI

AI Model : GPT-OSS 120B — Ope...

Build with AITalk to agent

Running in debug mode

You 9:31 PM

Create a monthly budget plan for a college student with ₹10,000 monthly expenses.

FinancialGuide_AI 9:31 PM | Show Reasoning

Here's a ready-to-use **Excel budget template** for a college student with a total monthly allowance of **₹10,000**.

How to use it

1. **Open the file** (download link above). It works in Excel, Google Sheets (via *File→Import*), or any other spreadsheet app.

2. **Enter your actual spending** each month in the **“Actual (₹)”** column.

3. The **“Difference (₹)”** column will automatically show how much you're under or over the planned amount (you can add a simple formula in your sheet: `=C2-B2` for each row).

4. Review the **“Savings / Emergency Fund”** row – aim to keep at least 10 % of your income (here ₹1,300) for unexpected expenses or future goals.

Budget breakdown (sample)

Category	Budgeted Amount (₹)	Actual (₹)	Difference (₹)

Type something...

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View all

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Build with AITalk to agent

Running in debug mode

Category	Budgeted Amount (₹)	Actual (₹)	Difference (₹)
Rent / Hostel	3,000		
Food (Home-cooked & Mess)	2,500		
Transport (Bus / Bike fuel)	800		
Study Materials (Books, Printouts)	600		
Internet / Mobile Data	500		
Entertainment & Socialising	600		
Clothing & Personal Care	500		
Savings / Emergency Fund	1,300		
Miscellaneous	200		
Total	10,000		

Type something...

Prompt 2 Answer Screenshot :

IBM watsonx Orchestrate

EnglishContact SalesBuy Now30 trial days left

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Running in debug mode

Miscellaneous	200
Total	10,000

Tips for staying on track

- Track daily – a quick note on your phone (e.g., a simple notes app) helps avoid “forgot-to-record” expenses.
- Review weekly – compare actual vs. budget; adjust categories if needed (e.g., you may need more for transport during exam weeks).
- Prioritise savings – treat the savings line as a non-negotiable expense, just like rent.
- Use digital payment records – UPI/IMPS transaction history can be exported and copied into the sheet for accurate actuals.

If you'd like a printable PDF version of this plan or have any specific categories you want to add, just let me know!

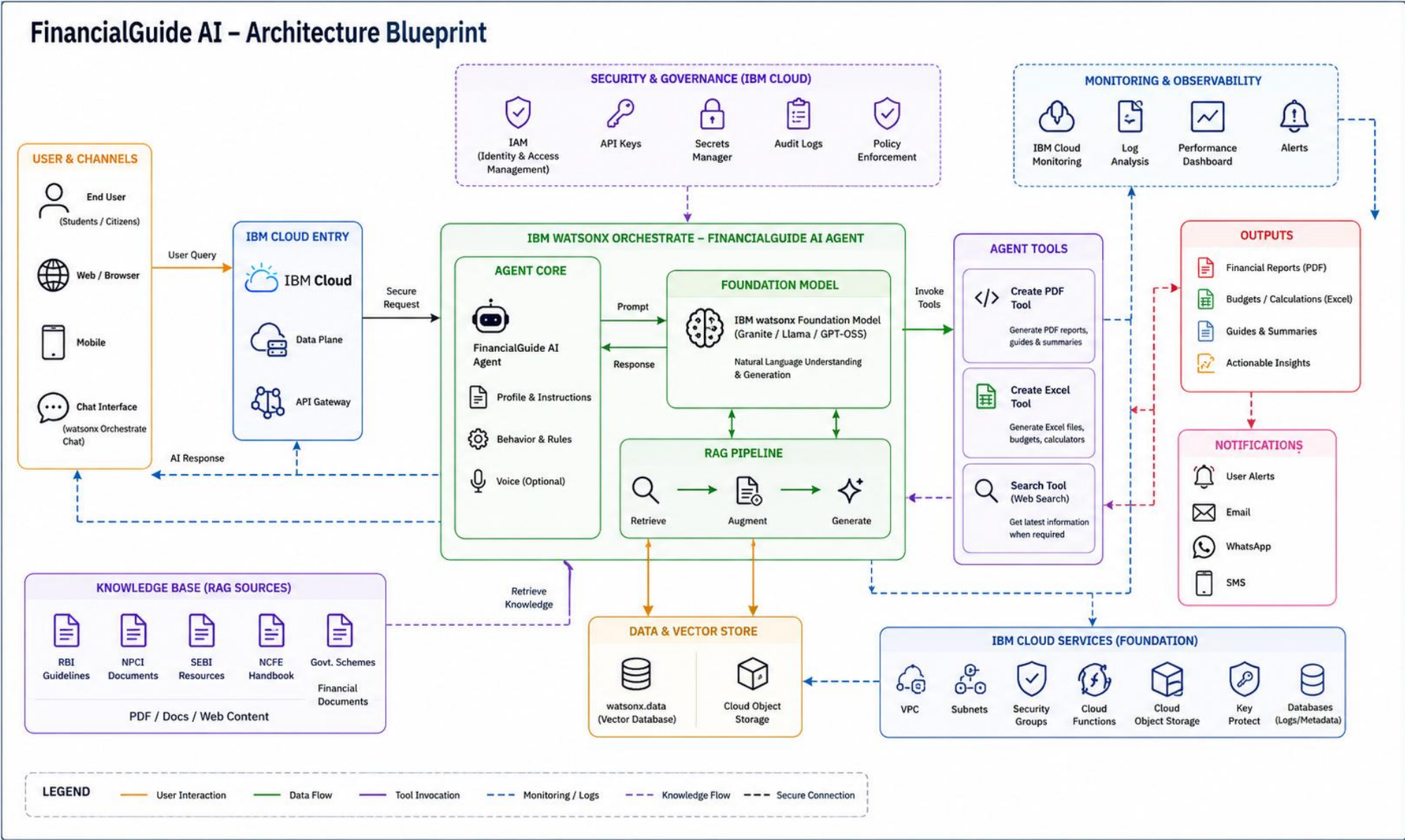
output.xlsx

Type something...

Prompt 2 Generated Excel Sheet Screenshot :

[illegible]

Architecture blueprint

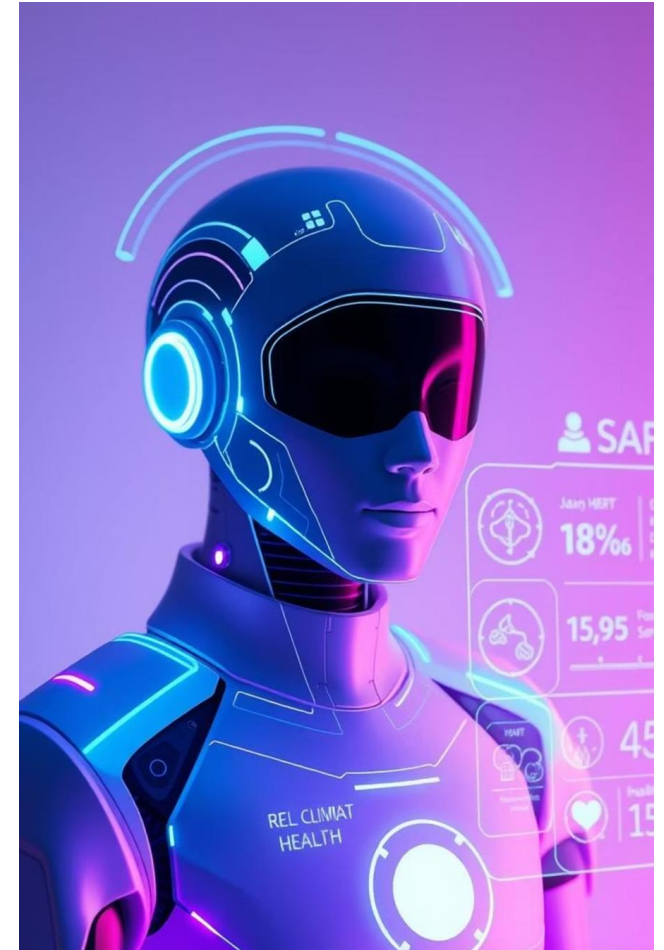


Role of Agentic AI in the solution

Agentic AI enables **FinancialGuide AI** to function as an intelligent, goal-oriented assistant rather than a traditional chatbot. Built using **IBM watsonx Orchestrate** and **IBM Granite Foundation Models**, the agent understands user queries and provides accurate, context-aware financial guidance.

The agent retrieves information from trusted knowledge sources such as **RBI, NPCI, SEBI, and NCFE** using **Retrieval-Augmented Generation (RAG)**, ensuring reliable and up-to-date responses. It assists users with topics including **UPI, digital payments, budgeting, savings, loans, investments, credit scores, and online fraud prevention**.

Agentic AI also supports intelligent decision-making by adapting its responses based on the user's questions and generating practical resources such as **PDF financial guides** and **Excel budget planners**. By combining reasoning, knowledge retrieval, and automation, FinancialGuide AI delivers a personalized, secure, and interactive financial learning experience, helping users make informed financial decisions with confidence.



Novelty and Uniqueness

The proposed **FinancialGuide AI** stands out by combining **Agentic AI**, **Retrieval-Augmented Generation (RAG)**, and **trusted financial knowledge** into a single intelligent assistant.

- **Agentic AI Intelligence** – Uses IBM watsonx Orchestrate to create an intelligent, goal-oriented financial assistant that provides contextual and interactive guidance.
- **RAG-Based Accuracy** – Retrieves information from trusted sources such as **RBI, NPCI, SEBI, and NCFE**, ensuring reliable and up-to-date financial advice.
- **Beginner-Friendly Guidance** – Explains complex financial concepts like UPI, SIP, loans, credit scores, and budgeting in simple, easy-to-understand language.
- **Financial Fraud Awareness** – Educates users about phishing, OTP scams, fake UPI requests, QR code fraud, and other cyber threats while promoting safe digital banking practices.
- **Interactive Productivity Tools** – Generates **PDF financial guides** and **Excel budget planners** to help users apply financial knowledge in real-life situations.
- **Scalable & Deployment-Ready** – Built on **IBM Cloud, IBM watsonx.ai, and IBM watsonx Orchestrate**, making it secure, scalable, and easy to extend with additional financial services and multilingual support.

This makes FinancialGuide AI a smart, reliable, and interactive digital financial assistant that empowers users to make informed financial decisions rather than acting as a simple question-answering chatbot.

Future Scope

1. Multilingual Support & Voice-Based Financial Assistant

The system can be enhanced with **multilingual support** and **AI-powered voice interaction**, enabling users to ask financial questions in their preferred language through voice or text. This will improve accessibility and make digital financial education more inclusive for users from diverse linguistic and educational backgrounds.

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2. Integration with Banking & Government Financial Services

Future versions can integrate with **banking APIs** and **government financial platforms** to provide real-time information on account services, loan eligibility, interest rates, government schemes, and personalized financial insights. This will transform FinancialGuide AI from an educational assistant into a comprehensive digital financial advisor while maintaining secure and reliable access to trusted financial information.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Kaustubh Dhakite

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 23, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/79aaf264-1d8b-4fa0-8235-f05f965c1488>



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In recognition of the commitment to achieve
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Kaustubh Dhakite

Has successfully satisfied the requirements for:

Getting Started with Cybersecurity



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Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Kaustubh Dhakite

for the completion of

Exploring Quantum Computing

(ALM-COURSE_3825253)

According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 50 mins

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Kaustubh Dhakite

Has successfully satisfied the requirements for:

Journey to Cloud: Envisioning Your Solution



Issued on: Jun 23, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/fb70427e-db42-44a7-984d-c9528a172929>



Certificates Earned:

IBM BOB Certificate

IBM SkillsBuild	Completion Certificate
	
This certificate is presented to	
Kaustubh Dhakite	
for the completion of	
Lab: Troubleshoot Your Code Using IBM Bob	
(ALM-COURSE_4071307)	
According to the Adobe Learning Manager system of record	
Completion date: 23 Jun 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.